The Weil-restriction isogeny census

Which curves over $\mathbb{F}_{q^n}$ an $\mathbb{F}_q$-rational $(\ell, \ldots, \ell)$-isogeny kernel in the Weil restriction reaches. Study `research/weil_restriction_isogeny_census_20261018/`; protocol `PROTOCOL.md`, frozen at `095c45c8a` with Amendment 1 at `d5bcdc081` before the first run; tool `examples/weil_restriction_isogeny_census.rs` (std-only Rust, 8 unit tests); recorded run `results/run-20261018T1605Z/` (Parts G and H, checks 1–9 pass). Second round of the Teske-variant search; round one is `research/teske_variants_field_types_20261018/` (PR #2170).

Labels: [T] proved or quoted from a source; [E] exact enumeration in the recorded run; [H] heuristic; [O] open. No cost is measured here; $\tilde{O}(q)$, $q^{3/2}$, $q^{4/3}$ are exponents.

Answer

The question was whether Weil restriction reaches curves that the known covering attacks do not. It does, and the reach can be written down exactly.

- **The association.** For $E/\mathbb{F}_{q^n}$ and a prime $\ell \neq p$, an $\mathbb{F}_q$-rational maximal isotropic subgroup $\Lambda \subset W[\ell]$ of the Weil restriction $W = \mathrm{Res}\, E$ that meets no conjugate factor $E^{\sigma^i}[\ell]$ (a *Tian kernel*) gives an $\mathbb{F}_q$-rational $(\ell, \ldots, \ell)$-isogeny $W \to W/\Lambda$ to a principally polarised abelian variety. For $n = 3$ Tian (J. Cryptology 2023) turns W/Λ into the Jacobian of a genus-3 curve in $\tilde{O}(\ell^3)$ and solves the DLP there in $\tilde{O}(q)$ (Diem–Thomé), against $q^{3/2}$ for rho and $\tilde{O}(q^{4/3})$ for Gaudry's decomposition; when $\#E$ is prime the curve is a plane quartic. Tian *assumes* Λ and gives one example ($q = 523$, $\ell = 3$, prime order); no source read states when Λ exists [T, §1].
- **Theorem 4 [T, E].** For $n = 3$ the existence of a Tian kernel depends only on $(q \bmod \ell, t \bmod \ell)$ and the type of $\varphi = \pi_{q^3}|E[\ell]$: it exists iff $\chi_\varphi(T) = T^2 - tT + q^3$ has a root in $\mathbb{F}_\ell$ and every root is a cube in $\mathbb{F}_\ell$, except that a Jordan φ (double root, φ not scalar) needs $\ell \equiv 1 \pmod 3$ as well. So nonsplit φ: never; split $\mathrm{diag}(\lambda, \mu)$: iff λ is a cube (automatic when $\ell \not\equiv 1 \bmod 3$); scalar λI: always; Jordan $J(\lambda)$: iff $\ell \equiv 1 \pmod 3$. The census confirms every cell at $\ell \in \{2, 3, 5, 7\}$ and every $q \bmod \ell$ (`kernels.csv`, 83 classes), and the split, scalar and nonsplit cases are proved for all ℓ (§3).
- **Which curves (Part H) [E].** Over $\mathbb{F}_{q^3}$, $q \in \{3, 5, 7, 11, 13\}$, every isomorphism class (56, 252, 690, 2662, 4400 of them) was classified at every $\ell \in \{2, 3, 5, 7\} \setminus \{p\}$. Prime-order classes with a Tian kernel at some odd $\ell \leq 7$: 0/8, 0/18, 24/61, 57/150, 144/323; all classes: 28/56, 156/252, 372/690, 1984/2662, 2714/4400. The prime-order curves reached are exactly those of Theorem 4 with no eigenvalue 1: at $q = 13$ the 135 prime-order curves with $t \equiv 2 \pmod 5$ ($\varphi = \mathrm{diag}(3, 4)$ on $E[5]$) and the 12 with $t \equiv 1 \pmod 3$ and $\varphi = -I$ on $E[3]$; at $q = 11$ the 54 with $t \equiv 0 \pmod 5$, the 3 with $\varphi = -I$ on $E[5]$ and the 12 with $\varphi = J(-1)$ on $E[7]$; at $q = 7$ the 24 with $t \equiv 1 \pmod 5$ and the 3 with $\varphi = -I$ on $E[3]$. At $q = 3, 5$ no admissible class exists for $\ell \leq 7$ ($q \bmod \ell$ forbids it), so the count is 0, as Theorem 4 predicts.
- **Covers never reach them [T, E].** At $\ell = 2$ a Tian kernel exists only for $\varphi = I$ (check 1), i.e. full rational 2-torsion, $\#E \equiv 0 \pmod 4$: the exact statement behind the cover attacks' "three rational points of order 2" (Diem 2003; Momose–Chao 2009; Tian–Li–Wang–Yu 2018), and the curves with an $\ell = 2$ kernel in Part H are precisely the 9, 42, 115, 443, 734 with full 2-torsion (check 7). Prime-order curves have no kernel at $\ell = 2$ (check 8) and are reached only through odd ℓ.
- **Assessment.** This is a *different type of affected curve* in the sense of the search: a congruence class, $(q \bmod \ell, t \bmod \ell)$ plus the splitting type of $E[\ell]$, containing prime-order curves, which no covering attack reaches and which the sources read do not name. It is **not hideable**: the test is $t \bmod \ell$ and the factor pattern of the ℓ-division polynomial, and the affected fraction is a constant (margin $a < 0$ in round one's framework). The attack path is Tian's; this round adds the existence theorem, the exact counts, and the $\ell = 2$ identification with the cover attacks. Two provisos are taken from Tian and not recomputed here: that W/Λ is geometrically indecomposable when Λ meets no factor (then a Jacobian by Oort–Ueno), and the $\tilde{O}(\ell^3)$ isogeny algorithm [T, O].
- $n = 2$: a Tian kernel exists iff φ^{-1} has a square root ψ with $\det \psi = -\bar{q}^{-1}$ (Amendment 1, check 2, unit test); the target is a genus-2 Jacobian, where index calculus costs $\tilde{O}(q)$ = rho, so nothing is gained. $n = 4$: kernels exist for 2, 8, 18 of 3, 10, 28 classes at $\ell = 2, 3, 5$; whether W/Λ is a Jacobian is the Schottky question and is left open [O].

1. Question, and what the sources say

Round one enumerated the *covering* associations (GHS, gGHS, Kummer compositums, the hyperelliptic genus-3 census) and proved (Theorem 3 there) that no hyperelliptic genus-3 Jacobian is isogenous to the Weil restriction of a prime-order curve over $\mathbb{F}_{q^3}$. Tian's introduction states the same boundary for covers [T]: "For $n = 3$, if we require C to be of genus 3, then a necessary condition is that the elliptic curve has three rational points of order 2 [3,13]" and "[19] shows that C needs to be non-hyperelliptic if the elliptic curve has prime order" (arXiv 2012.07173, §1; [3] Diem 2003, [13] Momose–Chao 2009, [19] Tian–Li–Wang–Yu, DCC 2018). His method: "an $\mathbb{F}_q$-rational (ℓ,ℓ,ℓ)-isogeny from the Weil restriction … to the Jacobian variety of a genus three curve over $\mathbb{F}_q$", solving "the discrete logarithm problem in some elliptic curve groups of prime order over $\mathbb{F}_{q^3}$ in a time of $\tilde{O}(q)$" (abstract). Section 3 opens with "Assume that we have an $\mathbb{F}_q$-rational maximal isotropic subgroup Λ of $W[\ell]$ … In order for the quotient W/Λ to be the Jacobian variety of a curve C, Λ should not contain any subgroups of E, E^σ or E^{σ^2}", and the hyperelliptic decision is by theta constants: "If 35 of them are non-zero, then C is hyperelliptic (this can happen only if the order of $E(\mathbb{F}_{q^3})$ is even). If 36 … non-hyperelliptic, … bound to happen if E has prime order." The one example is $q = 523$, $\mathbb{F}_{523^3} = \mathbb{F}_{523}[w]$, $w^3 + 2 = 0$, $y^2 = x^3 + (82w^2 + 92w + 140)x + 359w^2 + 339w + 243$, $\#E = 143069629$ prime, $\Lambda \subset W[3]$. No existence condition, no dependence on $q \bmod \ell$ or on the trace, and no proportion appear in the paper (the ar5iv text was read in full for these points). The web searches of this round (IACR ePrint, Springer, arXiv; queries recorded in the session) found no 2024–2026 follow-up stating an existence condition.

So the open question was concrete: **for which $E/\mathbb{F}_{q^3}$ does a Tian kernel exist at all?** It is a finite question about $(\ell, q \bmod \ell, [\varphi])$, answered here.

2. Setting (as frozen in the protocol)

$\varphi = \pi_{q^n}|E[\ell] \in GL_2(\mathbb{F}_\ell)$, $\det \varphi = q^n$, $\mathrm{tr}\, \varphi = t$, all mod ℓ. $W[\ell] = V = V_0 \oplus \ldots \oplus V_{n-1}$, $V_i = E^{\sigma^i}[\ell]$ identified with $E[\ell]$ through π_q^i; the $\mathbb{F}_q$-Frobenius acts as $\pi(v_0, \ldots, v_{n-1}) = (\varphi v_{n-1}, v_0, \ldots, v_{n-2})$, so $\pi^n = \varphi$ on each factor. The ℓ-Weil pairing of the Weil-restricted principal polarisation is $\omega_V = \oplus_i \bar{q}^i\, \omega$ ($\bar{q} = q \bmod \ell$, ω the standard form on $\mathbb{F}_{\ell^2}$; the Frobenius raises μ_ℓ to the q-th power), and $\omega_V(\pi v, \pi w) = \bar{q}\, \omega_V(v, w)$ (checked in the protocol's derivation and used by the tool). A **kernel** is an $\mathbb{F}_\ell[\pi]$-submodule of dimension n on which ω_V vanishes; a **Tian kernel** meets no V_i. As an $\mathbb{F}_\ell[T]$-module ($T = \pi$), $V = \mathrm{Ind}(E[\ell])$: the induced module of the $\mathbb{F}_\ell[S]$-module $E[\ell]$, $S = \varphi$, along $S = T^n$. This is the whole structure; q enters only through $\bar{q}$ and $\det \varphi$.

3. Theorem 4: Tian kernels for $n = 3$

Theorem 4. Let $\ell \neq p$ be prime and $E/\mathbb{F}_{q^3}$. A Tian kernel in $W[\ell]$ exists

- never, if $\chi_\varphi = T^2 - tT + q^3$ is irreducible over $\mathbb{F}_\ell$ (nonsplit);
- iff λ is a cube in $\mathbb{F}_\ell$, if $\varphi \sim \mathrm{diag}(\lambda, \mu)$ with $\lambda \neq \mu$ (then μ is a cube too, since $\lambda\mu = \bar{q}^3$);
- always, if $\varphi = \lambda I$;
- iff $\ell \equiv 1 \pmod 3$, if $\varphi \sim J(\lambda)$ (Jordan).

Proof of the nonsplit case [T]. `E[ℓ]` is a cyclic `F_ℓ[S]`-module, so `V ≅ F_ℓ[T]/(χ_φ(T^3))` is cyclic and its submodules are the monic divisors of `χ_φ(T^3) = T^6 − tT^3 + q^3`. A dimension-3 submodule needs a divisor of degree 3. The roots of `χ_φ(T^3)` are the cube roots of the two roots `α, ᾱ ∈ F_{ℓ^2} ∖ F_ℓ` of `χ_φ`; a root `c` in `F_{ℓ^3}` would have `c^3 ∈ F_{ℓ^3} ∩ F_{ℓ^2} = F_ℓ`, impossible, and a root in `F_ℓ` likewise. So `χ_φ(T^3)` has no irreducible factor of degree 1 or 3, hence no divisor of degree 3: there is no submodule of dimension 3 at all, let alone a kernel. (The tool asserts `submodules_dim_n = 0` on every nonsplit row; unit test `nonsplit_has_no_dim3_submodule`.)

Proof of the split case [T]. `V = A ⊕ B` with `A = Ind L_λ ≅ F_ℓ[T]/(T^3 − λ)`, `B = Ind L_μ ≅ F_ℓ[T]/(T^3 − μ)`, coprime, so every submodule is `A' ⊕ B'`. `A ∩ V_0 = L_λ` generates `A`, so a proper submodule `A'` meets `V_0` trivially, and the Tian kernels are the `A' ⊕ B'` with `(dim A', dim B') ∈ {(1, 2), (2, 1)}`. Submodules of dimension 1 or 2 of `F_ℓ[T]/(T^3 − λ)` exist iff `T^3 − λ` has a root `c ∈ F_ℓ`, i.e. iff `λ` is a cube. If it has, the eigenline `A' = ⟨(e, c^{−1}e, c^{−2}e)⟩` (`e ∈ L_λ`, `πa = ca`) satisfies `ω_V(a, πb) = (q̄/c) ω_V(a, b)`, so `A'^⊥ ∩ B` is a π-stable subspace of dimension 2 on which π has the quotient eigenvalue `d = q̄/c`, and `d^3 = q̄^3/λ = μ`: it is the degree-2 submodule of `B`. `B` is totally isotropic (`ω(L_μ, L_μ) = 0`), so `A' ⊕ (A'^⊥ ∩ B)` is a Tian kernel; by symmetry so is the `(2, 1)` one. Conversely without a cube root there is no `A'` of dimension 1 or 2. The count is 2 per cube root of `λ`: 2 when `ℓ ≢ 1 (mod 3)`, 6 when `ℓ ≡ 1 (mod 3)` and `λ` is a cube (both seen in `kernels.csv`: `ℓ = 3, 5` give 2, `ℓ = 7`, `diag(1, 6)` gives 6). Note `λμ = q̄^3` makes `λ` a cube iff `μ` is.

Scalar case [T for existence, E for the count]. `λ^2 = q̄^3` forces `ord(λ) | gcd(2(ℓ−1)/3, ℓ−1) = (ℓ−1)/3` when `ℓ ≡ 1 (mod 3)`, so an admissible scalar is always a cube and `T^3 − λ` has a root; `V = M^2` with `M = F_ℓ[T]/(T^3 − λ)`, and the split construction applied to any two independent eigenlines of π in different copies gives a Tian kernel. The counts `6, 12, 30, 72` at `ℓ = 2, 3, 5, 7` are the census's.

Jordan case [T for the negative half, E for the positive]. `V ≅ F_ℓ[T]/((T^3 − λ)^2)` is cyclic; `λ` is a cube as above. If `ℓ ≠ 1 (mod 3)`, `T^3 − λ = (T − c)h` with `h` irreducible quadratic and the only degree-3 divisor is `T^3 − λ` itself, whose submodule is `Ind L` for the eigenline `L ⊂ E[ℓ]`: it meets `V_0`. So no Tian kernel, as `kernels.csv` shows at `ℓ = 2, 3, 5`. If `ℓ ≡ 1 (mod 3)`, `T^3 − λ` splits, the seven degree-3 divisors give seven submodules, three are isotropic and two meet no factor (`ℓ = 7`, every admissible `q̄`; unit test `jordan_seven_divisors`). That the positive half holds for every `ℓ ≡ 1 (mod 3)` is [H] beyond `ℓ = 7`.

Corollary (prime order). A prime-order `E/F_{q^3}` has no eigenvalue 1 at any `ℓ`, so its Tian kernels come from: `φ = λI` with `λ ≠ 1` (`λ^2 = q̄^3`; e.g. `φ = −I` on `E[ℓ]`, `t ≡ −2 (mod ℓ)`), split `φ` with both eigenvalues cubes and `≠ 1`, or `φ = J(λ)`, `λ ≠ 1`, at `ℓ ≡ 1 (mod 3)`. Since split-versus-nonsplit is decided by whether `T^2 − tT + q^3` has a root mod `ℓ`, **for `ℓ ≢ 1 (mod 3)` the condition is purely a congruence on `(q, t) mod ℓ`** (plus "not Jordan" at a double root).

4. Part G: the census over the Frobenius classes [E]

`kernels.csv`: one row per `(n, ℓ, q̄, [φ])`, 83 rows at `n = 3`. Every cell of the frozen range ran (`n = 2, 3, 4`; `ℓ ≤ 7`, `ℓ ≤ 5` at `n = 4`; 75 s). Checks: 1 (`n = 3`, `ℓ = 2`: kernels only for `φ = I`): 0 violations; 2 (`n = 2`: every Tian kernel is a graph with `det ψ = −q̄^{−1}`, `φψ^2 = I`): 0 violations; 3 (P2): at `(3, 3, 1)` the class `φ = 2I` (no eigenvalue 1) has 12 Tian kernels, which is where Tian's `q = 523 ≡ 1 (mod 3)`, `ℓ = 3` example lives; 4 (representative independence, two alternative representatives per split/nonsplit/Jordan class): 0 mismatches.

`(n, ℓ)`	classes `[φ]` (all $\bar{q}$)	with a Tian kernel	… and no eigenvalue 1
(2, 2)	3	2	1
(2, 3)	10	5	2
(2, 5)	28	14	9
(2, 7)	54	27	20
(3, 2)	3	1	0
(3, 3)	8	3	1
(3, 5)	24	10	6
(3, 7)	48	15	6
(4, 2)	3	2	1
(4, 3)	10	8	5
(4, 5)	28	18	10

`n = 3`, the rows with a Tian kernel (all others have none):

ℓ	$\bar{q}$	class	`tr`	`det`	eigenvalue 1	submodules (dim 3)	kernels	Tian kernels
2	1	scalar `I`	0	1	yes	15	9	6
3	1	scalar `2I`	1	1	no	40	16	12
3	1	scalar `I`	2	1	yes	40	16	12
3	2	split `(1,2)`	0	2	yes	4	4	2
5	1	split `(2,3)`	0	1	no	4	4	2
5	1	scalar `I`	2	1	yes	156	36	30
5	1	scalar `−I`	3	1	no	156	36	30
5	2	split `(2,4)`	1	3	no	4	4	2
5	2	split `(1,3)`	4	3	yes	4	4	2
5	3	split `(3,4)`	2	2	no	4	4	2

5	3	split (1,2)	3	2	yes	4	4	2
5	4	split (1,4)	0	4	yes	4	4	2
5	4	scalar 3I	1	4	no	156	36	30
5	4	scalar 2I	4	4	no	156	36	30
7	1, 2, 4	scalar I	2	1	yes	560	80	72
7	1, 2, 4	Jordan J(1)	2	1	yes	7	3	2
7	1, 2, 4	scalar −I	5	1	no	560	80	72
7	1, 2, 4	Jordan J(−1)	5	1	no	7	3	2
7	3, 5, 6	split (1,6)	0	6	yes	20	8	6

The nonsplit rows have 0 submodules of dimension 3 (Theorem 4). The $\ell = 7$ split classes (2,4), (3,5) (det 1) and (2,3), (4,5) (det 6) have exactly the two submodules A, B and no Tian kernel: 2, 3, 4, 5 are not cubes mod 7 (the cubes are 1, 6). The split rows at $\ell = 3, 5$ are every split class there is: every element is a cube.

5. Part H: every curve over F_{q^3} [E]

curves.csv (4400 + 2662 + 690 + 252 + 56 = 8060 classes, one row per (curve, ℓ)), summary.csv, by_class.csv, checks_h.txt. Check 5 (class count) is exact at every q; checks 6–8 have 0 violations; the class of φ was identified for every (curve, ℓ) (class_not_found = 0).

q	classes	prime order	cofactor 2	full 2-torsion (= kernel at $\ell = 2$)	kernel at some odd $\ell \le 7$: all	prime order
3	56	8	7	9	28	0
5	252	18	25	42	156	0
7	690	61	48	115	372	24
11	2662	150	150	443	1984	57
13	4400	323	288	734	2714	144

By ℓ (with_tian, prime-order, cofactor-2); $\ell = p$ is excluded:

q	$\ell = 3$	$\ell = 5$	$\ell = 7$
3	–	26, 0, 6	8, 0, 0
5	124, 0, 0	–	40, 0, 0
7	58, 3, 6	342, 24, 0	–
11	1330, 0, 0	710, 57, 78	758, 12, 12
13	368, 12, 24	2198, 135, 114	732, 0, 0

The classes behind the prime-order counts (by_class.csv), each a cell of Theorem 4 without eigenvalue 1: q = 7: $\ell = 3$, φ = 2I (3 curves of 29 in the class); $\ell = 5$, $\bar{q} = 2$, diag(2,4), t ≡ 1 (24 of 171). q = 11: $\ell = 5$, $\bar{q} = 1$, $\varphi = -I$ (3 of 22) and diag(2,3), t ≡ 0 (54 of 666); $\ell = 7$, $\bar{q} = 4$, J(−1), t ≡ 5 (12 of 370). q = 13: $\ell = 3$, φ = 2I (12 of 184); $\ell = 5$, $\bar{q} = 3$, diag(3,4), t ≡ 2 (135 of 1099). At q = 5 no admissible class without eigenvalue 1 exists for $\ell \le 7$ (q ≡ 2 mod 3 gives only diag(1,2) at $\ell = 3$, q ≡ 5 mod 7 only diag(1,6) at $\ell = 7$), and at q = 3 the one such class, $\ell = 5$, $\bar{q} = 3$, diag(3,4), holds 13 curves of which none has prime order (6 have cofactor 2): the zeros at q = 3, 5 are the absence or emptiness of a cell, not counter-examples to Theorem 4.

Reading. (i) The curves reached at odd ℓ are a constant fraction at every q (50–75% of all classes, 37–45% of the prime-order classes once a non-empty admissible cell exists), and the fraction moves with q mod ℓ, not with q: at q = 11 the $\ell = 3$ cell is the eigenvalue-1 class diag(1,2) (no prime-order curve), at q = 13 it is 2I (12 prime-order curves). (ii) Cofactor-2 curves are reached at odd ℓ too (per ℓ, summary.csv: 6 at q = 3; 6 at q = 7; 78 + 12 at q = 11; 24 + 114 at q = 13, a curve with kernels at two ℓ counted twice): the "cofactor 2" question of Tian–Li–Wang–Yu has an isogeny-side answer. (iii) full_2_torsion = with_tian($\ell = 2$) at every q and check 7 has no violation: the $\ell = 2$ Tian kernels are exactly the full-2-torsion curves, the necessary condition of the genus-3 cover attacks (Diem; Momose–Chao), and the compositum covers of round one (Theorem 2 there) live inside this set, consistent with Joux–Vitse's "about half of the curves having their full 2-torsion".

6. What is affected, at what cost, and whether it can hide

- **Affected set (n = 3).** Prime-order E/F_{q^3} with (q mod ℓ, t mod ℓ) in a cell of Theorem 4 without eigenvalue 1, for some small $\ell \neq p$. For $\ell = 3$: q ≡ 1 (mod 3) and t ≡ 1 (mod 3) with $\varphi = -I$ (the 3-torsion field is F_{q^6}). For $\ell = 5$: q ≡ 1: t ≡ 0 (split) or t ≡ 3 (−I); q ≡ 2: t ≡ 1; q ≡ 3: t ≡ 2; q ≡ 4: t ≡ 1 or 4 with φ scalar. For $\ell = 7$: q ≡ 1, 2, 4 and t ≡ 5 (−I or J(−1)). Larger ℓ add cells; heuristically [H] a random prime-order curve has a cube-split φ at a given ℓ with probability about 1/2 · 1/3 ($\ell \equiv 1 \bmod 3$) or 1/2 ($\ell \equiv 2 \bmod 3$) minus the eigenvalue-1 cases, so almost every prime-order curve is reached at some $\ell \le 20$; the exact small-ℓ figures above are the evidence.
- **Cost.** Transfer Õ(ℓ^3) [T, Tian] plus the Jacobian DLP Õ(q) [T, Diem–Thomé], against q^{3/2} (rho) and Õ(q^{4/3}) (Gaudry's decomposition, which applies to every curve over F_{q^3}). Exponents only; the constants of Diem–Thomé's algorithm are not measured here [O].
- **Auditor's test.** Compute t, check the congruences, and factor the ℓ-division polynomial over F_{q^3} for the splitting type (Part H does exactly this; seconds at toy size, polynomial time in general). The affected set is a constant fraction of every admissible trace class, so in round one's margin framework a < 0: nothing to hide, nothing to walk to; the isogeny class decides, up to the scalar/Jordan distinction at a double root (an ℓ-volcano-level property, also polynomial-time).

- **What the census does not prove.** That W/Λ is a Jacobian (Tian's indecomposability condition Λ ∩ V_i = 0 is what we counted; Oort–Ueno then gives a Jacobian or its Serre twist for a geometrically indecomposable 3-dimensional ppav), and that the explicit isogeny is as cheap as stated: both are Tian's [T], not re-derived; no isogeny was computed in this round [O].

7. n = 2 and n = 4

n = 2 [T, E]: Amendment 1's closed form is confirmed (check 2; the unit test n2_closed_form compares the census with a brute-force search over GL_2(F_ℓ) at ℓ = 3, 5): a Tian kernel exists iff φ^{−1} = ψ^2 for some ψ with det ψ = −q̃^{−1}. Classes with a kernel and no eigenvalue 1: 1, 2, 9, 20 of 3, 10, 28, 54 at ℓ = 2, 3, 5, 7. The target is a 2-dimensional ppav, a genus-2 Jacobian or a Weil restriction of an elliptic curve over F_{q^2} again; genus-2 index calculus costs Õ(q), the same as rho on E(F_{q^2}), so n = 2 is "reached" without being weaker (as Scholten's and Joux–Vitse's genus-2 covers already are).

n = 4 [E, O]: kernels exist for 2, 8, 18 of the 3, 10, 28 classes at ℓ = 2, 3, 5 (1, 5, 10 without eigenvalue 1). W/Λ is a 4-dimensional ppav; a Jacobian only on the Schottky locus (codimension 1 in A_4), and if a non-hyperelliptic genus-4 Jacobian, Diem's index calculus would cost Õ(q^{4/3}) against q^2 (rho) and Õ(q^{3/2}) (Gaudry). Whether any W/Λ is a Jacobian is open; it is the one place in this round where a *sparse* affected set could exist (a 1/q Schottky fraction of the kernels), and the next step is explicit: at q ≤ 13, ℓ = 3, construct W/Λ by Tian's theta method generalised to dimension 4 and test the Schottky–Igusa modular form.

8. Checks and controls

check	where	result
1. n = 3, ℓ = 2: Tian kernels only for φ = I (P3)	checks_g.txt	pass (0 violations)
2. n = 2: every Tian kernel is a graph with det ψ = −q̃^{−1}, φψ^2 = I (Amendment 1)	checks_g.txt, unit test	pass (0 violations; census = brute force at ℓ = 3, 5)
3. P2: a class without eigenvalue 1 has a kernel at (3, 3, 1)	checks_g.txt	pass (φ = 2I, 12 kernels)
4. representative independence	checks_g.txt	pass (0 mismatches)
5. class counts 2(q^3 − 2) + gcd(4, q^3 − 1) + gcd(6, q^3 − 1) (p ≥ 5); 2(q^3 − 1) + 4 at q = 3	checks_h.txt	pass at every q
6. division polynomials: degree, squarefree; cubic degree 3	checks_h.txt	pass (0 violations)
7. ℓ = 2 class I iff the cubic splits iff full 2-torsion; 4 | #E then; kernel at ℓ = 2 only then	checks_h.txt, summary.csv	pass (full_2_torsion = with_tian)
8. prime-order curves: no eigenvalue 1 at any ℓ, no kernel at ℓ = 2	checks_h.txt	pass
9. prime-order fractions reported per q and ℓ	summary.csv, §5	reported
Hasse bound; class found for every (curve, ℓ)	checks_h.txt	pass
unit tests (8): rref, n = 2 closed form, orbit patterns, division polynomials at q = 11, 2-torsion vs order at q = 7, class count at q = 5, Jordan ℓ = 7 divisors, nonsplit has no dim-3 submodule	rustc --test	pass

Predictions: P1 as corrected (Amendment 1) holds; P2 holds; P3 holds; P4 holds in the form "a constant fraction once an admissible cell exists" (0 at q = 3, 5 is the absence of a cell, see §5).

9. What is not established

- [O] No isogeny computed, no DLP solved, no constant measured. The Jacobian property of W/Λ and the Õ(ℓ^3) isogeny are Tian's.
- [H] The Jordan positive half of Theorem 4 beyond ℓ = 7; the heuristic density of cube-split classes at larger ℓ.
- [O] n = 4: the Schottky question.
- [O] Characteristic 2 base fields in Part H (the table fields are odd); Part G is characteristic-free.
- [O] No Lean statement for Theorem 4 (its nonsplit and split halves are short module-theory arguments; a formalisation is owed under AGENTS.md).
- [O] Conductor could not be installed (installer HTTP 403 through the session proxy, as in round one); no task card.
- Nothing here concerns a standardised curve; F_{q^3} fields are the literature's setting (Tian, Joux–Vitse, Momose–Chao).

10. Canonical graphs and diagram

- figures/mechanism.svg/png: the isogeny route, Theorem 4's decision table, and the cover route for contrast (figures/diagram.py).
- figures/affected.svg/png: fraction of classes over F_{q^3} with a Tian kernel, all and prime-order, by q, and the prime-order fraction by ℓ (summary.csv).
- figures/kernel_map.svg/png: n = 3, Tian kernels by (ℓ, q mod ℓ, [φ]) (kernels.csv).
- figures/n2n4.svg/png: classes with a kernel by (n, ℓ) (kernels.csv).
- reports/run-20261018T1605Z/run-report.{typ,pdf}: environment, log, checks, the summary and the n = 3 kernel table, the graphs.
- REPORT.typ/pdf from this file (build.sh).

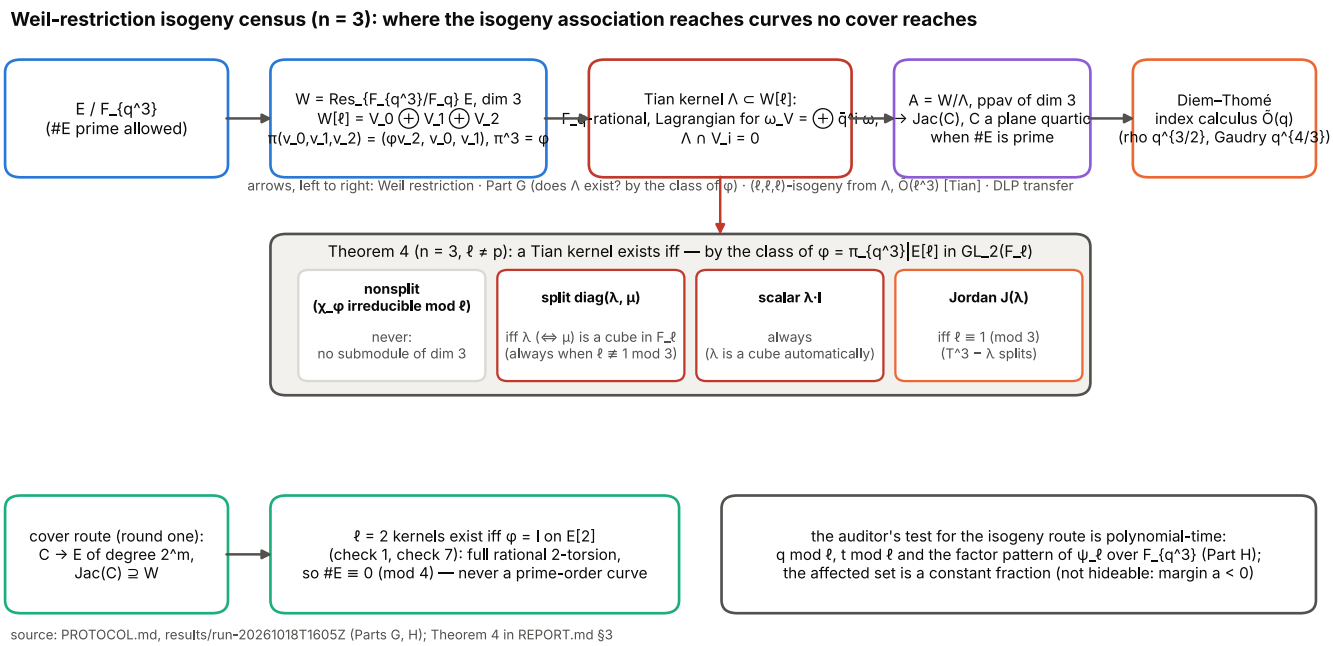
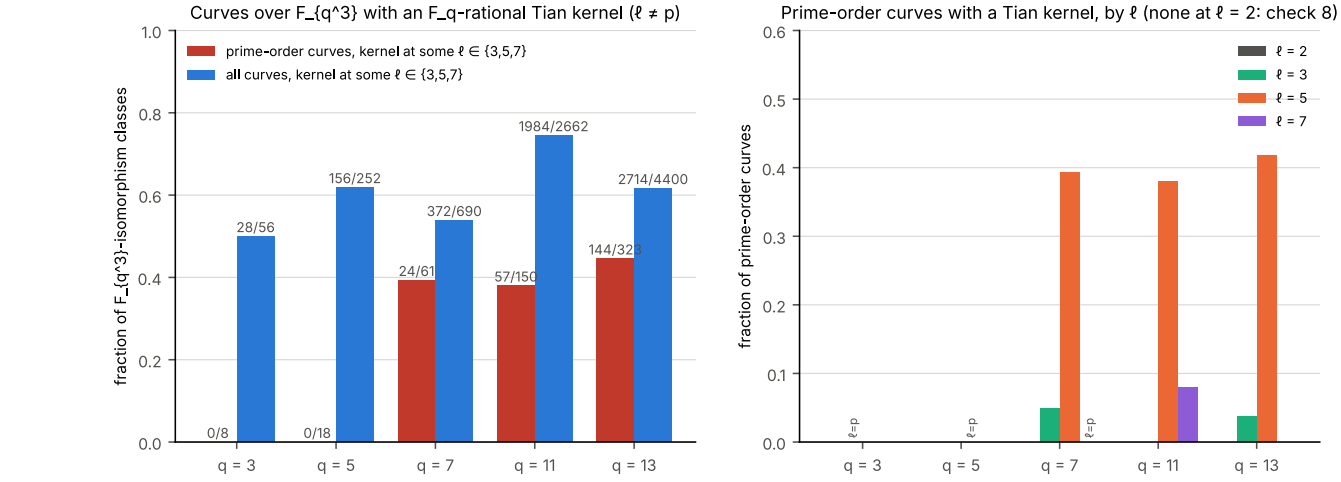
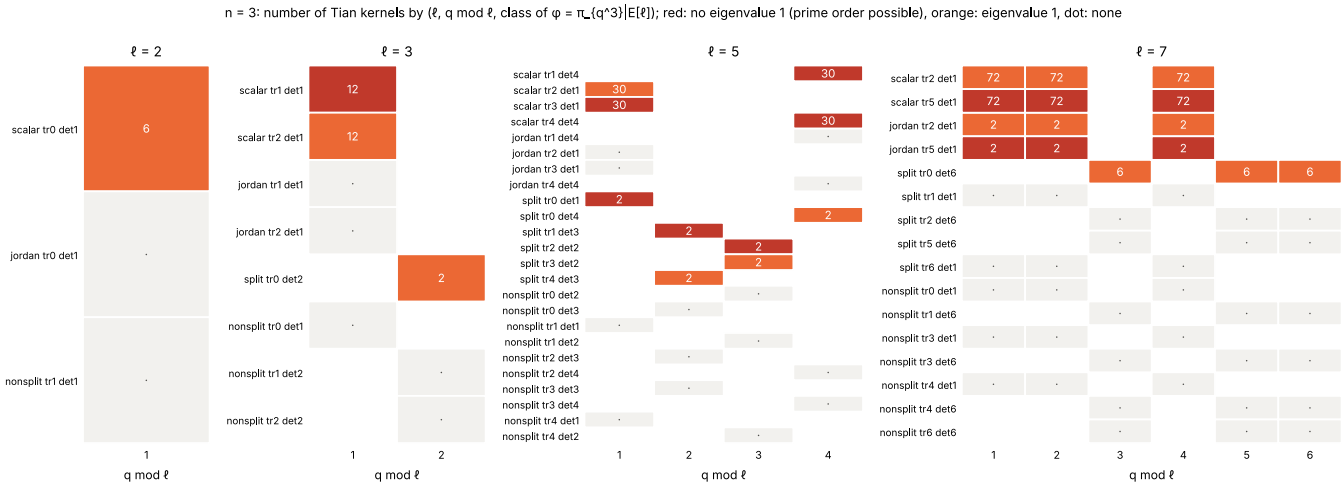


Figure 1: The isogeny route, Theorem 4's decision table, and the cover route (figures/mechanism.svg).



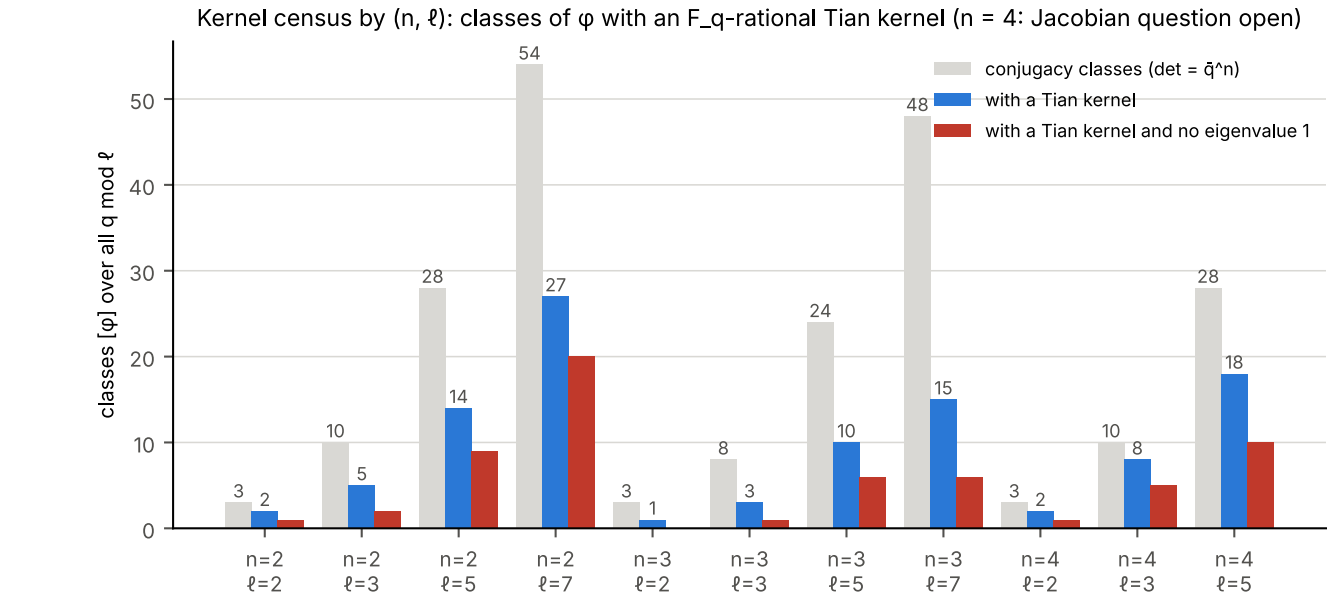
source: run-20261018T1605Z/summary.csv (exact counts)

Figure 2: Curves over `F_{q^3}` with a Tian kernel, by `q` and by ℓ (figures/affected.svg; `summary.csv`).



source: run-20261018T1605Z/kernels.csv (exact; one row per conjugacy class)

Figure 3: `n = 3`: Tian kernels by `(ℓ, q mod ℓ, [φ])` (figures/kernel_map.svg; `kernels.csv`).



source: run-20261018T1605Z/kernels.csv

Figure 4: Kernel census by `(n, ℓ)` (figures/n2n4.svg; `kernels.csv`).

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- Round one: `research/teske_variants_field_types_20261018/REPORT.md` (Theorems 2 and 3, Part B's `(3,5)`/`(2,6)` types, Part C's census).